Modern Thermodynamics

RandomWalks2005

1 Basic Concepts and The Laws of Gasses

1.1 Thermodynamic Systems

Note; Later in these notes I use N and n interchangably. This is bad practice, as these are not the same thing. This mistake is not catastrophic for now, but keep note of it. There is also some slight conflation between what is called natural units of temperature and laboratory units of temperature. In certain places, I do omit the Boltzmann constant...

The fundamental thermodynamic act is that of distinction. When we speak about different thermodynamic environments, we have to speak about the interior of that environment (the system) and the exterior of that environment (the environment).

For a system or an environment to be well defined we require a boundary which is also well defined to differentiate between them. In this sense, we can see the boundary as preceding both the system and its environment.

We use this boundary so that we can speak about how the system and its environment relate to one another in terms of matter and energy. From this, a classification of 3 different types of systems emerge:

1. Isolated Systems: Do not exchange energy or matter (Idealization).
2. Closed systems: Exchange energy but not mass.
3. Open systems: Exchange both energy and mass (e.g. biological systems).

Mathematically thermodynamics involves the use of multi-variable calculus because thermodynamic quantities are typically functions of multiple variables which describe the state or configuration of a system. There are two main types of variables:

1. Intensive variables: Size-independent (e.g Pressure, Temp)
2. Extensive variables: Size-dependent (e.g. Volume, moles)

1.2 Thermodynamic Equilibrium

The evolution of a thermodynamic system tends to a specific state called thermal equilibrium. This tendency is irreversible, similar to the irreversibility of spontaneous chemical reactions. A nonequilibrium state can be defined as a configuration for a system where irreversible processes are still acting.

The Zeroth law of thermodynamics is that thermal equilibrium is transitive: If A is in equilibrium with B and B is in equilibrium with C then A and C are in equilibrium. In equilibrium systems the state variables are globally defined, but this is not true of nonequilbrium systems; These are only defined locally.

1.3 Biological and Other Open Systems

Biological systems are interesting because they are an example of the fact that open systems can demonstrate spontaneous organization. It is possible per the theory of dissipative structures for entropy of an open system to be exported to its environment so that the net local entropy increase is negligible.

1.4 The Laws of Gases

We define the conversion between various scales for measuring temperature:

$$T(C) = (5/9)[T(F) - 32],\ \ T(K) = T(C) + 273.15 \tag{1}$$

We define the gas constant, the Boltzmann constant, and the Avogadro number, all of which will be crucially important:

$$R = 8.314\text{JK}(1/\text{mol}) = 0.08314\text{bar L }(1/\text{K})\ (1/\text{mol}) = 0.0821\text{atm L }(1/\text{K})\ (1/\text{mol}) \tag{2}$$

$$N_A = 6.023 \times 10^{23} 1/\text{mol} \tag{3}$$

$$k_B = R/N_A = 1.3807 \times 10^{-23}\text{J}(1/\text{K}) \tag{4}$$

Boyles law found that $V = \frac{f_1(T)}{P}$ at fixed T for $f_1(T)$ as some function of T (The volume of a gas is inversely proportional to its pressure).

Charles law found that at constant pressure volume and temperature are directly proportional: $\frac{V}{T} = f_2(p)$ for $f_2(p)$ as some function of pressure.

Avogadros law connected Boyles law to the number of moles: $V = N\frac{f_1(T)}{p}$

These previous laws lead to the law of ideal gasses:

$$pV = NRT \tag{5}$$

The ideal gas law describes gasses which do not have strong intermolecular forces or large molecular sizes involved in them, there are also some considerations to be made at high pressures which lead to modifications of the ideal gas law.

For a mixture of gasses we treat them as separable by Dalton's law of partial pressures, giving:

$$p_k V = N_k RT \tag{6}$$

Gay-Lussac found that during vacuum expansion, the internal energy is invariant despite changes in volume and pressure, implying that $U = U(T)$ only. Because U is extensive, it scales with size of system and thus scales proportional to N. Therefore:

$$U(T, N) = NU_m(T) \tag{7}$$

With U_m being the total internal energy per mole. Total energy for a mixture of gasses is the sum of its energies. In other words, the separability of gasses holds.

Eventually U_m was defined more concretely as:

$$U_m = cRT + U_0 \tag{8}$$

With U_0 as a constant and c changes with the atomicity of the gas e.g c=3/2 for monatomic, 5/2 for diatomic. c comes from kinetic molecular theory.

Gay-Lussac also showed that at constant pressure relative changes in volume due to increases in temperature had constant values across all dilute gasses, which lead the foundation for volume displacement temperatures, as they captured the relation: $V = V_0(1 + \alpha t)$, with $\alpha = 1/273$, representing the coefficient of expansion at constant pressure.

The first edit of the ideal gas law from van der Waals:

$$(p + a\frac{N^2}{V^2})(V - Nb) = NRT \tag{9}$$

Which propagates to the energy as:

$$U_{vw} = U_{ideal} - a(\frac{N}{V})^2 V \tag{10}$$

Attempt at deriving the modified ideal gas law: We start by thinking about an ideal gas, which we note as having restricted molecular size and perfectly elastic collisions. The first deviation which we may notice is an increased amount of container volume occupied by the gas in nonideal scenarios. We can let some constant b represent the excluded gas and note that this b is an extensive property. In other words, it seems to scale with the number of molecules in the system, the adjusted volume of this term is given by $V_{adjusted} = (V_{container} - Nb)$

Now we must think about how the effects of this volume change will propagate to the pressure. We know that in ideal circumstances collisions are perfectly elastic, which also means that the flux of molecules through the boundary which contains the gas is relatively uniform; However, we can imagine a scenario in which a molecule is pulled away from the boundary prior to its collision by intermolecular forces. By the separable nature of force: $P_{adjusted} = P_{container} + P_{corrective}$. The strength of this corrective pressure should depend on the amount of molecules not colliding with the boundary as well as the amount of molecules colliding with the boundary, both of which would be proportional to the mole/volume density, meaning that this corrective pressure would be proportional to $(n/v)^2$. We can go further to say that there must exist some constant which expresses the ratio of non-colliding to colliding molecules and we let this constant of proportionality be a such that $P_{corrective} = a(n/V)^2$. Now we substitute the P and the V from $PV = NRT$ and it yields the final adjusted van der waals equation. We can obtain even more information about these constants through dimensional analysis.

1.5 States of Matter and the Van Der Waals Equations

The simplest type of transformation in thermodynamics is caused by heat melting solids and vaporizing liquids. Solid, liquid, and gas refer to something called a phase of matter. Each compound has temperature T_m at which it melts from a solid to a liquid, and a temperature T_b at which it boils (at standard pressure). Interestingly, this has implications for statistical mechanics by the relationship $T = 1/\beta$, since it implies the existence of a unique beta at the melting and boiling points.

At the melting or boiling point, heat does not necessarily result in an increase in temperature, only in the conversion of one phase to another. This heat that does not increase the temperature is called the latent heat.

Key limitation of ideal gas: Does not explain the conversion of gas to liquid upon compression.

Caginard de la tour: Gas does not liquefy when compressed unless its temperature is below a special value called the critical temperature. Van der waals linked this idea to molecular volume and molecular forces. Further modifying the ideal gas law:

$$p = \frac{NRT}{V - bN} - \delta p \tag{11}$$

Van der waals related δp to N/V using the kinetic theory of gasses to obtain the familiar revised ideal gas law previously discussed, rewritten in its common form:

$$(p + a\frac{N^2}{V^2})(V - Nb) = NRT \tag{12}$$

The theoretical significance of this equation is the unification of liquids and gasses under the banner of fluids: Van der Waals equation successfully became a description for the behavior of these fluids rather than simply for gasses. We reveal this by looking at a plot of p-V for a given temperature (called the p-V isotherm). Three cases:

1. $T > T_C$: Resembles ideal gas isotherm (far from liquid).
2. $T < T_C$: Isotherm has a max and min, liquid and gas phase coexist

3. $T = T_C$: Distinction between gas and liquid disappears. Here we note a critical pressure p_c and a critical molar volume V_{mc}. These critical values are called the critical constants.

(See pp. 20 for more about discussion of critical temperature isotherms).

We can notice from the isotherms that for $T < T_C$ we have the derivative of pressure with respect to volume equal to zero, with second derivatives of opposite signs. At the critical temperature these second derivatives approach zero so that both the first derivative and second derivative are equal to zero.

This gives way to the relationship:

$$a = (9/8)RT_C V_{mc} \quad b = (V_{mc})/3 \tag{13}$$

With V_{mc} as molar critical value.

Solving for the critical parameters in terms of a and b we get:

$$T_C = (8a)/(27Rb) \quad p_C = (a)/(27b^2) \quad V_{mc} = 3b \tag{14}$$

Law of Corresponding States: Noting that gasses have critical parameters which characterize them and depend on certain molecular qualities, Van der Waals used this to introduce a dimensionless reduced version of the variables. The motivating factor here was to get an expression for the critical parameters independent of a and b. In general for a state variable X and its critical value X_C the reduced variable X_R is given by $X_R = X/X_C$.

Van der waals was then able to obtain:

$$p_r = \frac{8T_r}{3V_r - 1} - \frac{3}{V_r^2} \tag{15}$$

This implies that gasses have the same reduced pressure at a given value of reduced volume and reduced temperature. This essentially means that fluids act close to identical with scaled parameters.

From this a general idea of deviation from the ideal gas law was defined by compressibility:

$$Z = \frac{V_m}{V_{m,ideal}} = \frac{pV_m}{RT} \tag{16}$$

Ideal gas behavior is $Z = 1$. Z is predicted for most gasses at 3/8, but Van der waals noticed this was not true experimentally. All fluids have a specific temperature called the boyle temperature where they act as ideal fluids. We can literally interpret this as slow molecules feeling the stickiness of attractions and fast molecules slipping through the attractive forces.

Further revision to the ideal gas law based on Van der waals own criticism has led to many revisions but resulted in the commonly used Redlich-Kwong equation:

$$p = \frac{NRT}{V - Nb} - \frac{a}{\sqrt{T}} \frac{N^2}{V(V + Nb)} = \frac{RT}{V_m - b} - \frac{a}{\sqrt{T}} \frac{1}{V_m(V_m + b)} \tag{17}$$

a and b here are different from the a and b of the van der waals equation. Limitations of these kind of equations come from their two parameterization. Larger and more complex molecules exhibit more sophisticated IMFs and therefore would need additional parameters to adequately describe their complexity.

There is also a molecular view of the law of corresponding states:

The a and b parameters of the Van der waals equation describe characteristics of an intermolecular force called the van der waals force. Van der waals forces are repulsive at close distances and attractive at far distances. Two important consequences: - Repulsivity results in volume - Repulsivity makes it hard to condense or compress phases

The potential energy between molecules is expressed by the Lennard-Jones energy:

$$U_{LJ} = 4\epsilon[(\frac{\sigma}{r})^{12} - (\frac{\sigma}{r})^{6}] \tag{18}$$

See (pp 24 for a graphical exploration of the LJ energy). This model breaks down when we have to consider molecular orientations.

1.6 An introduction to kinetic theory of gasses

Through the ideas of force and stochastic motion it is possible to relate pressure (F/A) to force exerted across a given area by molecular collisions.

Assumptions: 1.) Moving in any direction is equally probable.

We take the average velocity and decompose it into x,y,z:

$$v_{avg}^2 = v_{xavg}^2 + v_{yavg}^2 + v_{zavg}^2 \tag{19}$$

By assumption 1:

$$v_{xavg}^2 = v_{yavg}^2 = v_{zavg}^2 = \frac{v_{avg}^2}{3} \tag{20}$$

Now we take N_A, N, V, M, m, n where M is the molar mass, m is the mass of a single molecule and n is the number of molecules per unit of volume.

Assume a gas chamber with thickness Δx. Given the perfectly elastic nature of the collision, it starts with momentum mv and ends with momentum -mv, making the change in momentum $2mv$ where we use the average velocity.

We can say that approximately half the molecules will be moving away from the wall and the other half towards the wall for a time $t = \Delta x / v_{avg}$. We can also say that in one layer we have $(\Delta x A n)/2$ molecules colliding with the wallls. We can say that the total momentum is the product of an individual momentum with the number of particles colliding with the walls which is equal to $2mv_{xavg}(\Delta x A n)/2$. Therefore we get:

$$F = \frac{Momentum}{\Delta t} = \frac{2mv_{xavg}\Delta x A n}{\Delta t}(n/2) = \frac{mv_{xavg}\Delta x A n}{(\Delta x / v_{xavg})} = mv_{xavg}^2 n A \tag{21}$$

We get pressure by simply dividing by the area which yields:

$$p = (1/3)mnv^2 = (1/3)M(N/V)v^2 \tag{22}$$

We are able to say from this that $RT = (1/3)Mv^2$ and that $(1/2)mv^2 = (3/2)k_BT$

We see from all of this as we increase atomicity we increase the internal molar energy because we are considering additional directions of motion (degrees of freedom)

Maxwell-Boltzmann Distribution

Pressure measurement does not tell us the fraction of molecules which have velocities of a particular magnitude and direction; Maxwell sought out to find this by looking into the velocity probability distribution for molecules. $P(v)dv_xdv_ydv_z$ is the fraction of the total number of molecules with velocity vectors with components in the range $(v_x, v_x + dv_x), (v_y, v_y + dv_y), (v_z, v_z + dv_z)$. Each point in velocity space corresponds to a velocity vector. $P(v)dv_xdv_ydv_z$ is the probability that the velocity of a molecule is within an elemental volume at the point $P(v)$. This $P(v)$ is called the probability density in velocity space. Boltzmann generalizes this to energy. If $\rho(E)$ is the number of different states in whcih the molecule has energy E:

$$P(E) \propto \rho(E)e^{-E/k_bT} \tag{23}$$

We call $\rho(E)$ the density of states and this previous proportionality is called the Boltzmann principle. We use this principle to derive equilibrium thermodynamical properties– This is statistical thermodynamics.

The energy of a molecule $E = E_{trans} + E_{rot} + E_{vib} + ...$ where these are different translational, rotational, vibrational and other forms of energy.

According to the Boltzmann principle:

$$P(v)dv_xdv_ydv_z = \frac{1}{z}e^{-mv^2/2k_BT}dv_xdv_ydv_z \tag{24}$$

where $v^2 = v_x^2 + v_y^2 + v_z^2$. z is the normalization factor and:

$$z = \int\int\int e^{-mv^2/2k_BT}dv_xdv_ydv_z \tag{25}$$

With each integral going from negative infinity to positive infinity. The normalization is defined in such a way such that dividing by z makes the distribution equal to 1, to satisfy the core tenet of probability theory.

We can calculate the normalization factor z as:

$$1/z = (\frac{m}{2\pi k_BT})^{3/2} \tag{26}$$

The normalized probability distribution for the velocity can then be rewritten as:

$$P(v)dv_xdv_ydv_z = (\frac{m}{2\pi k_BT})^{3/2}e^{-mv^2/2k_BT}dv_xdv_ydv_z \tag{27}$$

This is the maxwell velocity distribution for a gas at equilibrium. The width of the distribution is proportional to the temperature.

Maxwell Speed Distribution

Average velocity of a molecule is zero since the directions all cancel out in being equally probable, but since average speed depends on the magnitude of the velocity, it is not zero. Therefore, we can derive the speed distribution from the probability distribution.

We do this by integrating $P(v)$ over all directions where the velocity can point. In spherical coordinates, since the volume element is $v^2 \sin(\theta) d\theta d\phi dv$, the probability is written as $P(v)v^2 \sin(\theta) d\phi dv$. The integral is then:

$$\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} P(v)v^2 \sin(\theta) d\theta d\phi dv = 4\pi P(v)v^2 dv \tag{28}$$

That end quantity is the probability distribution for molecular speed. Using our value of P(v) obtained earlier, we can conclude that:

$$f(v)dv = 4\pi(\frac{m}{2k_B T})^{3/2} e^{-\beta v^2} v^2 dv \tag{29}$$

Where $\beta = m/(2k_B T)$

We can also rewrite this using the molar mass and the gas constant if we recall that. The average speed is given by: $[V] = \int_0^\infty v f(v) dv$

We use Mathematica or Maple to calculate these distributions. These programs handle them much better by condensing the form using 1/z and using beta instead of the full expanded form. We obtain the average speed as:

$$[V] = \sqrt{\frac{8RT}{\pi M}} \tag{30}$$

A similar treatment of average energy yields:

$$[\frac{1}{2}mv^2] = \frac{3}{2}k_B T \tag{31}$$

We can note that the most probable speed is calculated by maximizing $f(v)$.

SEE APPENDICES FOR A REFRESHER ON PARTIAL DERIVATIVES AND PROBABILITY THEORY AS WELL AS USEFUL INTEGRALS AND MATHEMATICA PROGRAMS